OAKLAND COMMUNITY COLLEGE®
ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2040 MIDTERM

NO FLASH DRIVES ARE ALLOWED

IF A FLASH DRIVE IS FOUND IN THE COMPUTER A ZERO GRADE WILL BE ISSUED.

SAVE BEFORE DOWNLOADING.

IF THE SYSTEM FAILS AND YOU DID NOT SAVE, YOU MAY LOSE YOUR WORK AND WILL HAVE TO REDO.

ALLOTTED TIME:

THREE (3) HOURS FROM THE COURSE SCHEDULED START TIME.

LADDER DIAGRAMS ARE GENERIC, NOT SCREEN CAPTURES

The following problems uses Studio 5000, Emulate 5000, and FactoryTalk SE Client

REMEMBER TO PRINT TO OneNote

Mid-term program for Studio 5000 one must use the named file: **PROG1.**
Points: 10
Approximate time for completion: 10 minutes

Mid-term program for Studio 5000 two must use the named file: **PROG2.**
Points: 15
Approximate time for completion: 20 minutes

Mid-term program for Studio 5000 three must use the named file: **PROG3.**
Points: 25
Approximate time for completion: 30 minutes

The following problems uses Siemens TIA PLC and HMI.

REMEMBER TO PRINT TO OneNote

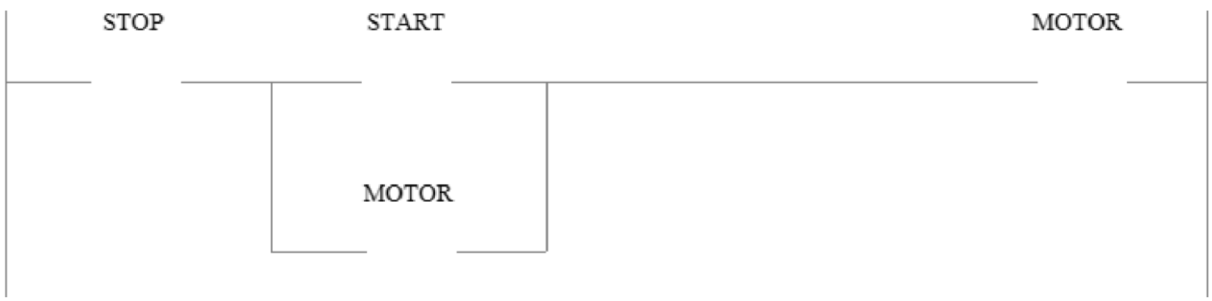
Mid-term program for Siemens TIA one must use the named file: **PROG1.**
Points: 10
Approximate time for completion: 10 minutes

Mid-term program for Siemens TIA two must use the named file: **PROG2.**
Points: 15
Approximate time for completion: 20 minutes

Mid-term program for Siemens TIA three must use the named file: **PROG3.**
Points: 25
Approximate time for completion: 30 minutes

Remember to print to OneNote in the MIDTERM Tab as PROG1

Create a program that when a normally open momentary START pushbutton is pressed, a motor will turn on. When the START pushbutton is released, the motor will continue running. Pressing a normally closed momentary STOP pushbutton will turn off the motor



INPUTS:

Address	Controller Tag	Description
[redacted]	[redacted]	Momentary Normally Open Pushbutton
[redacted]	[redacted]	Momentary Normally Closed Pushbutton

OUTPUT:

Address	Controller Tag	Description
[redacted]	[redacted]	Motor Starter

<u>SEQUENCE</u>

USE HMI **PROG1_HMI**

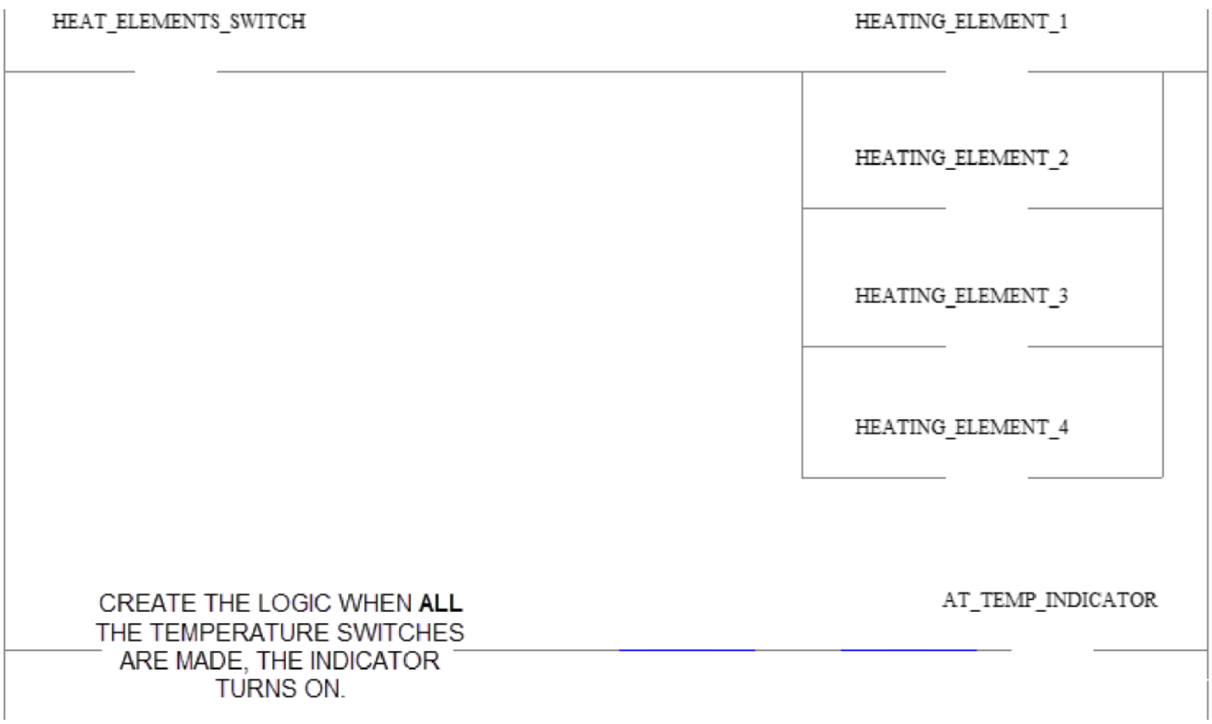
① Press the START pushbutton. MOTOR is ON.

② Press the STOP pushbutton. MOTOR is OFF

Remember to print to OneNote in the MIDTERM Tab as PROG1

Remember to print to OneNote in the MIDTERM Tab as PROG2

A machine has four heating elements that will allow operation when there is a temperature of 125 degrees at each heating element. The heating elements are turned on with a normally open maintained switch. <u>When a heating element is at temperature, a set of normally open contacts on the temperature switch for that heating element are made</u>. When **all** four heating elements are at temperature, that is all four temperature switches are closed, the AT TEMPERATURE indicator is turned on.



INPUTS:

Address	Controller Tag	Description
[redacted]	[redacted]	Power Switch Normally Open
[redacted]	[redacted]	Temperature Switch No. 1
[redacted]	[redacted]	Temperature Switch No. 2
[redacted]	[redacted]	Temperature Switch No. 3
[redacted]	[redacted]	Temperature Switch No. 4

OUTPUTS:

Address	Controller Tag	Description
[redacted]	[redacted]	Heating Elements At Temperature Indicator
[redacted]	[redacted]	Heating Element No. 1
[redacted]	[redacted]	Heating Element No. 2
[redacted]	[redacted]	Heating Element No. 3
[redacted]	[redacted]	Heating Element No. 4

<u>SEQUENCE</u>

USE HMI **PROG2_HMI**

① Turn the HEATING ELEMENTS SWITCH to ON.

HEATING ELEMENT 1 thru 4 are ON.

TEMP SWITCHES 1 thru 4 are made.

AT TEMP INDICATOR turns ON.

② Turn the HEATING ELEMENTS SWITCH to OFF.

HEATING ELEMENT 1 thru 4 turn OFF.

TEMP SWITCHES 1 thru 4 are apart.

AT TEMP INDICATOR turns OFF.

Remember to print to OneNote in the MIDTERM Tab as PROG2

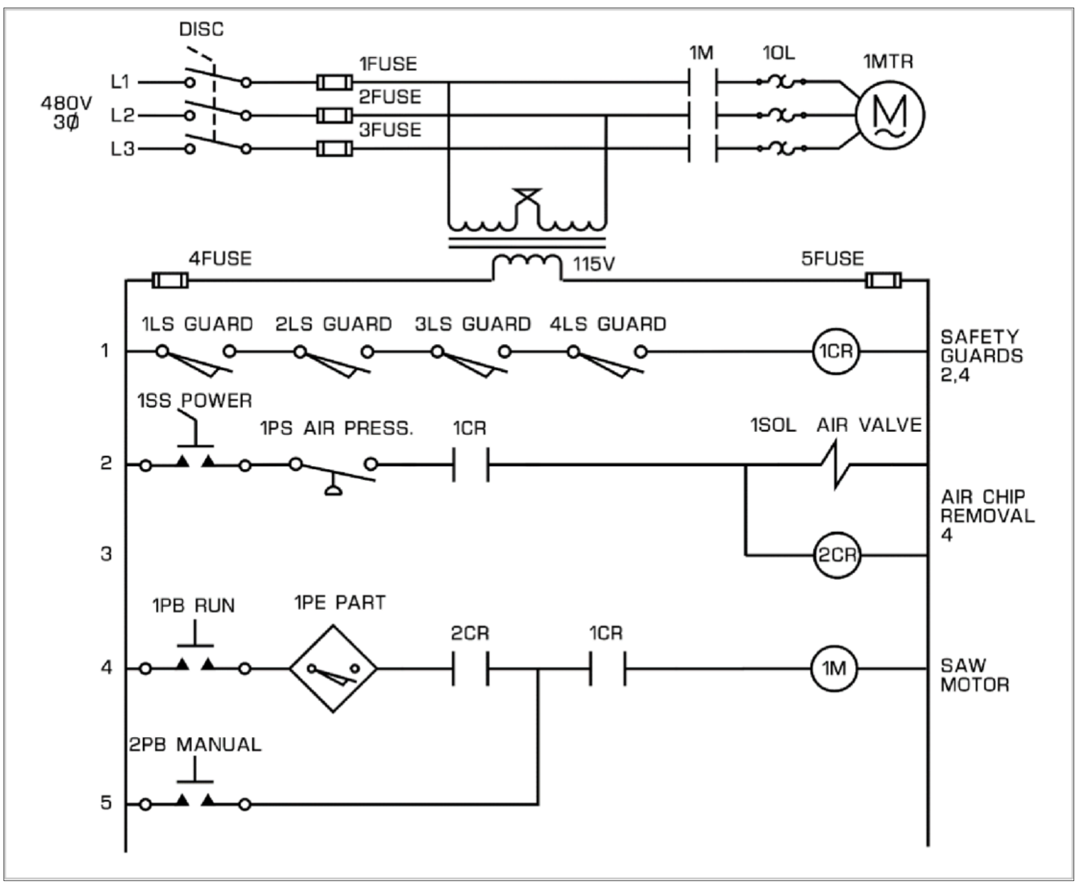
Remember to print to OneNote in the MIDTERM Tab as PROG3

Create the logic for the control of a saw.

- When **all** GUARDS are made, CR1 (Safety Guards) is energized.
- When 1SS (Power Selector Switch) is turned **and** 1PS (Pressure Switch) is made **and** 1CR (Safety Guards) has been energized; the 1SOL (Air Valve) and 2CR (Chip Removal) are energized.
- When 1PB (Run) **and** 1PE (Part) are made **and** 2CR (Chip Removal) has been energized **and** 1CR (Safety Guards) has been energized, **or** 2PB (Manual) is made **and** 1CR (Safety Guards) has been energized; 1M (Saw Motor) is energized.

All input switches, sensors, and contacts are wired **Normally Open** and ***actuated*** when output is to be ON .

Hardwired Circuit



SEE PAGE 6 FOR LOGIC STRUCTURE AND SEQUENCE.

INPUTS:

Address	Controller Tag	Description
[redacted]		1LS N.O. GUARD
		2LS N.O. GUARD
		3LS N.O. GUARD
		4LS N.O. GUARD
		1SS N.O. POWER
		1PS N.O. AIR PRESS.
		1PB N.O. RUN
		1PE N.O. PART
		2PB N.O. MANUAL

OUTPUTS:

Address	Controller Tag	Description
[redacted]		1SOL AIR VALVE
		1M SAW MOTOR

INTERNALS:

The Control Relays use Program Tags.

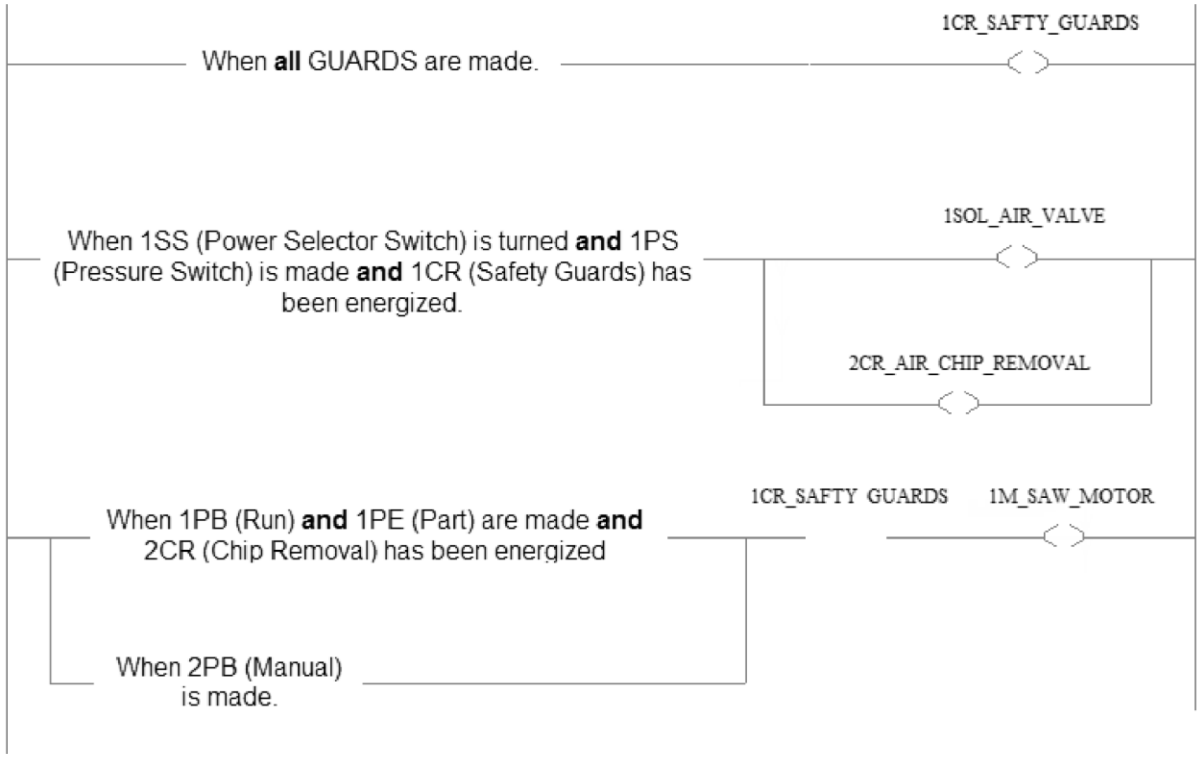
Parameter and Local Tag	Data Type	Description
[redacted]		1CR SAFETY GUARDS
		2CR AIR CHIP REMOVAL

CR = Control Relay
LS = Limit Switch
N.O. = Normally Open
PS = Pressure Switch

SOL = Solenoid
PB = Push Button
SS = Selector Switch, Maintained
PE = Photoelectric Switch

continue

PLC Program



<u>SEQUENCE</u>

USE HMI **PROG3_HMI**

① When PLC is placed in RUN MODE, the LIMIT SWITCHES for the GUARDS will be made. CONTROL RELAY 1 is energized.

② Turn to On the SELECTOR SWITCH at the HMI,.

③ The PRESSURE SWITCH will be made. The AIR VALVE and CONTROL RELAY 2 are energized.

④ Select the button on the HMI for the PART. PHOTO ELECTRIC 1 is made

⑤ Select and hold on the HMI the button for RUN. The SAW MOTOR is energized. When the RUN button is released the SAW MOTOR is de-energized.

⑥ Select on the HMI the button for GUARD 2 FAULT. The LIMIT SWITCH 2 is apart and CONTROL RELAY 1 is de-energized.

⑦ Select and hold on the HMI the button for RUN, the SAW MOTOR should not be energized.

⑧ Select the button on the HMI for NO PART and select the button for LIMIT SWITCH 2 to remove the fault. Turn to Off the SELECTOR SWITCH.

⑨ Select and hold on the HMI the button for MANUAL. The SAW MOTOR is energized. When the MANUAL button is released the SAW MOTOR is de-energized.

⑩ Select on the HMI the button for GUARD 2 FAULT. The LIMIT SWITCH 2 is apart and CONTROL RELAY 1 is de-energized. Press and hold the MANUAL button, the SAW MOTOR should not be energized.

Remember to print to OneNote in the MIDTERM Tab as PROG3

PROG1 SIEMENS

Remember to print to OneNote in the MIDTERM Tab as PROG1

Write a program that when a normally open SWITCH is turned, an EXHAUST FAN and INDICATOR LAMP will turn on.

Ignore any Blocks or Tags which names begin with Emulate.

INPUT:

Address	PLC Tag	Description
[redacted]		Maintained Normally Open Switch

OUTPUTS:

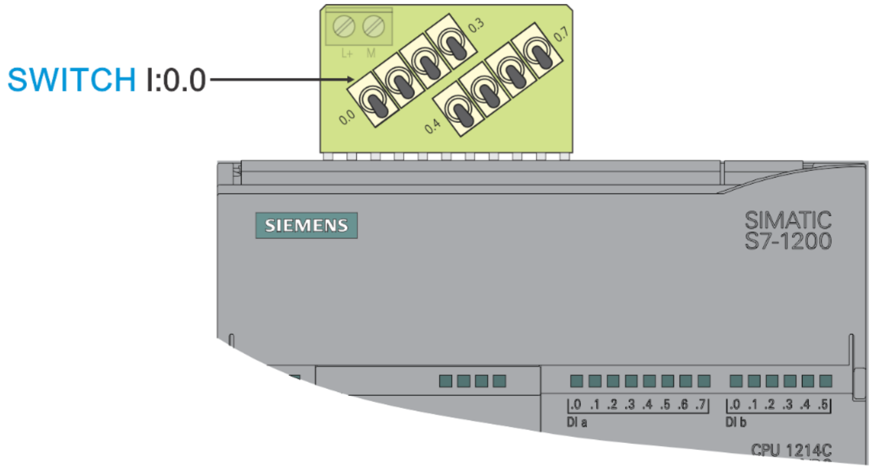
Address	PLC Tag	Description
[redacted]		Fan Motor
		Fan Running Indicator

SEE NEXT PAGE FOR SWITCH SEQUENCE

DOWNLOAD HMI FOR EXAM PROBLEM.

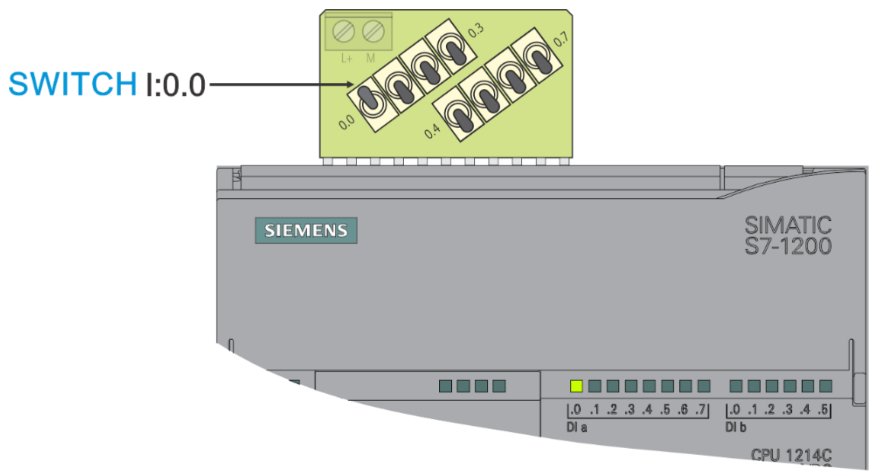
SEQUENCE

Initial switch settings,
push down **all** switches.



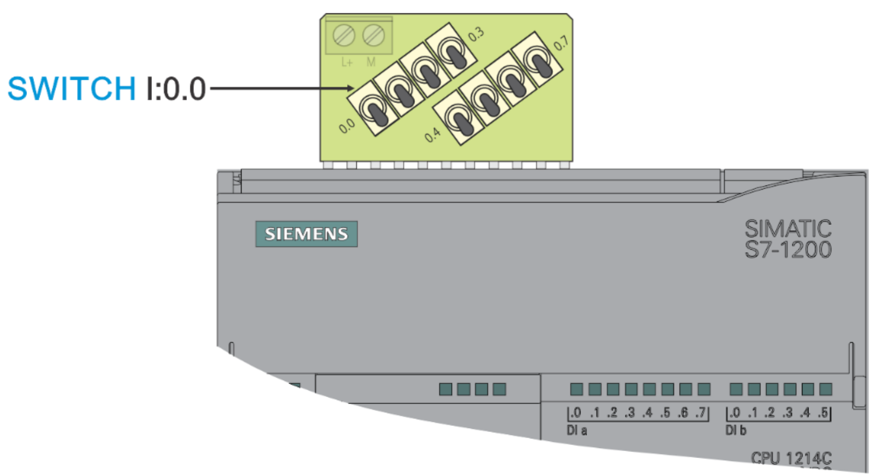
Push up **SWITCH**.

EXHAUST FAN and INDICATOR LAMP are ON.



Push down **SWITCH**.

EXHAUST FAN and INDICATOR LAMP are OFF.



Remember to print to OneNote in the MIDTERM Tab as PROG1

PROG2 SIEMENS

Remember to print to OneNote in the MIDTERM Tab as PROG2

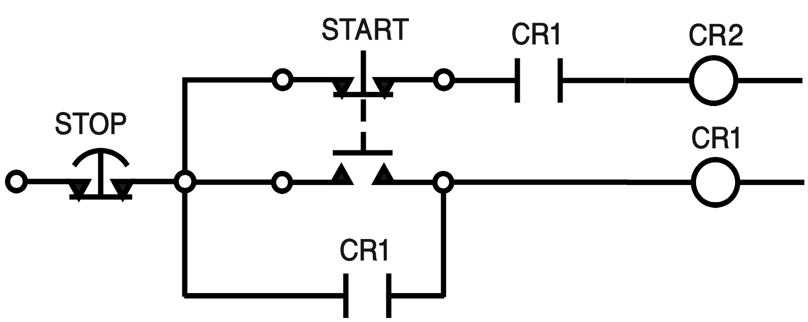
Convert the hardwired circuit below to a ladder logic program. This logic creates a circuit when the START is pressed an outputs turns On. Releasing the START pushbutton turns On a second output. Pressing and releasing STOP turns Off both outputs.

- When the START pushbutton is pressed, the output CR1 will be energized.
- Releasing the same START pushbutton then energizes CR2.
- The STOP pushbutton de-energizes both the outputs.

For the hardwired version the START is a double pole - double throw pushbutton with normally open and normally closed contacts.

SEE NOTE ON NEXT PAGE: START***** is a single Normally Open pushbutton for the PLC version of the circuit.

Hardwired Circuit



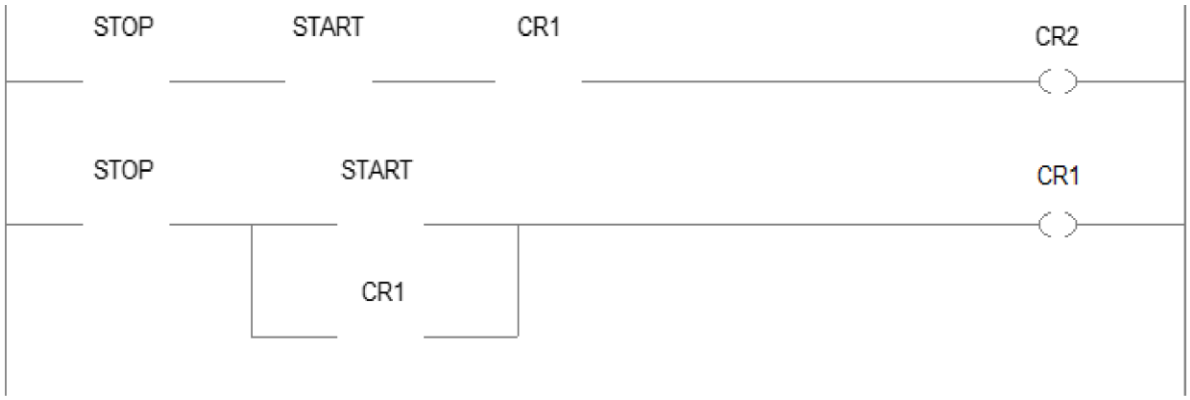
INPUTS:

Address	PLC Tag	Description
[redacted]	[redacted]	Momentary Normally Open Pushbutton
[redacted]	[redacted]	Momentary Normally Closed Pushbutton

OUTPUTS:

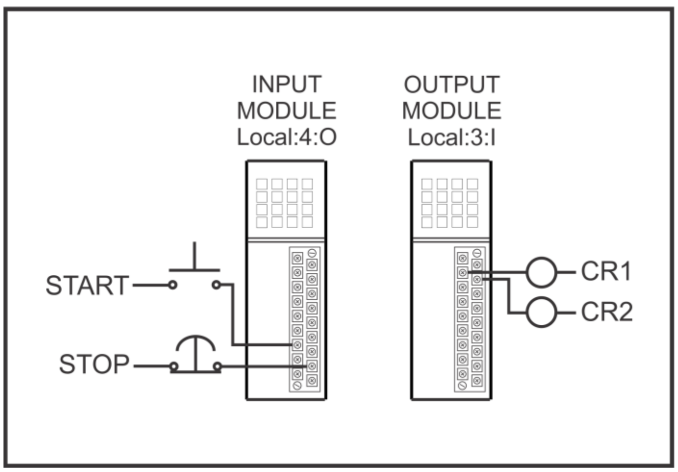
Address	PLC Tag	Description
[redacted]	[redacted]	First output on
[redacted]	[redacted]	Second output on

PLC Program

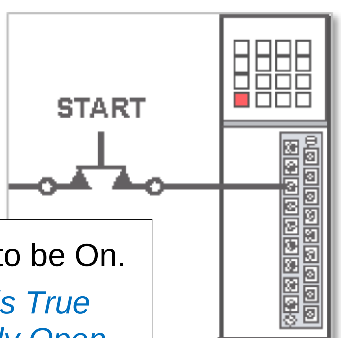


NOTE: START is a single Normally Open pushbutton for the PLC version of the logic.

PLC INPUT/OUTPUT INTERFACE FOR ABOVE CIRCUIT

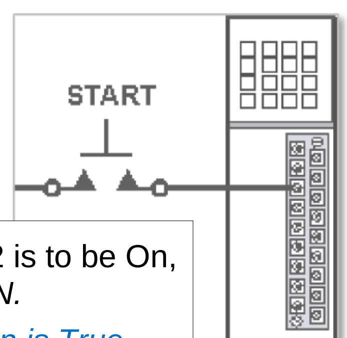


STATUS OF START PUSHBUTTON WHEN:



Pressed, CR1 is to be On.

*What instruction is True when the Normally Open pushbutton is **actuated**.*



Released, CR2 is to be On, *after CR1 is ON.*

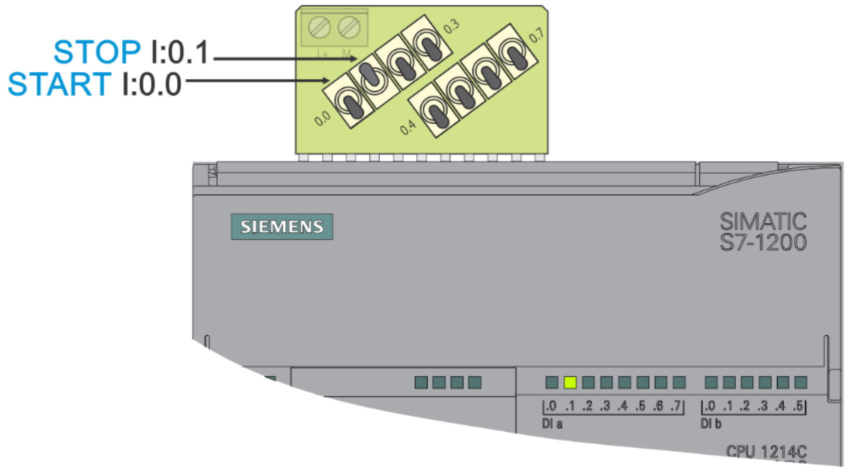
*What instruction is True when the Normally Open pushbutton is **unactuated**.*

DOWNLOAD HMI FOR EXAM PROBLEM.

<u>SEQUENCE</u>

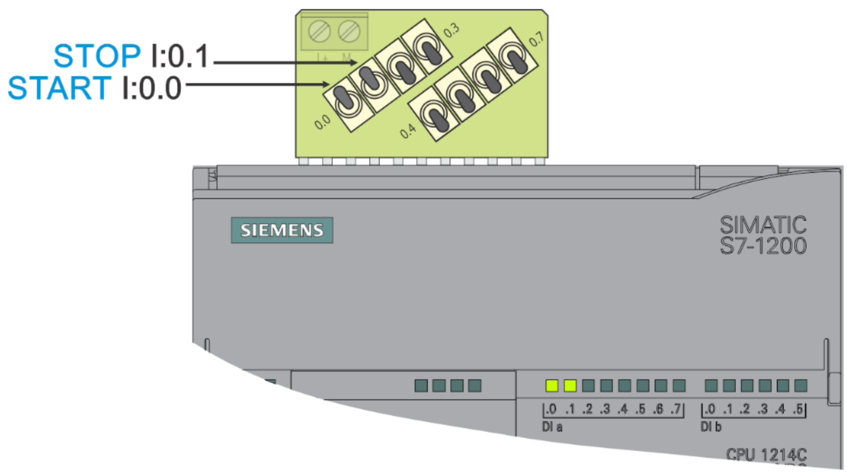
Initial switch settings, push up **STOP** switch.

All the *other* switches are down.



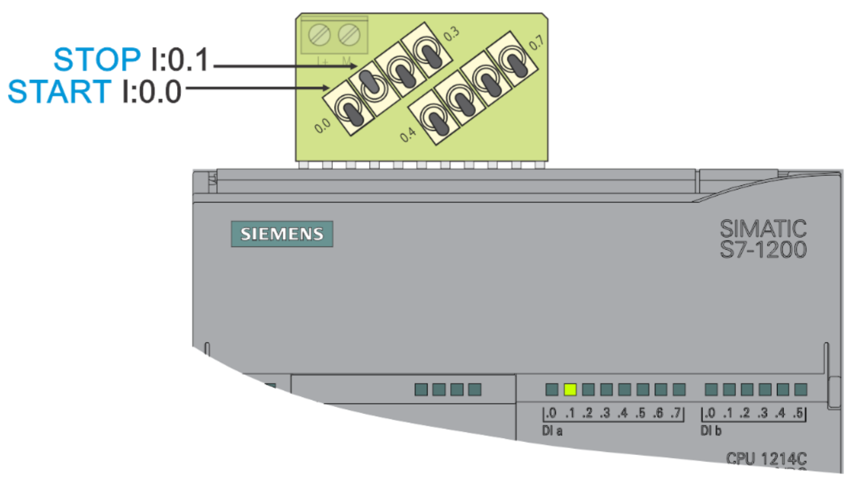
Push up the **START** switch.

CR1 is ON.



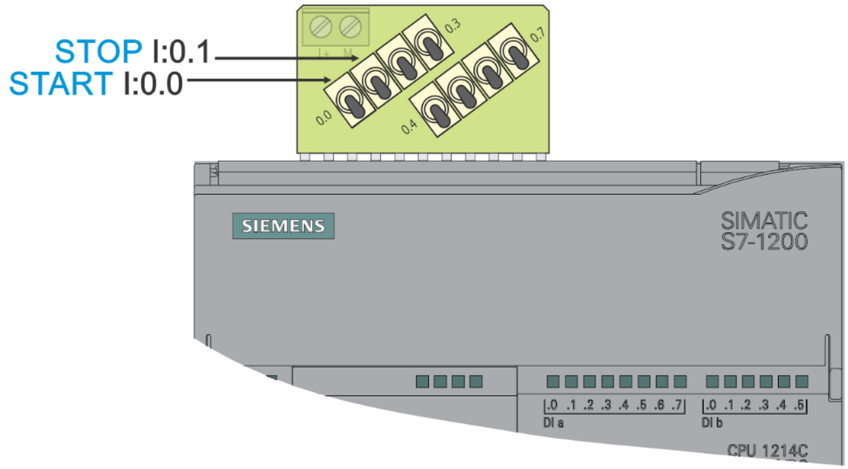
Push down the **START** switch.

CR2 is ON.



Push down the **STOP** switch.

CR1 and CR2 are OFF.



Remember to print to OneNote in the MIDTERM Tab as PROG2

PROG3 SIEMENS

Remember to print to OneNote in the MIDTERM Tab as PROG3

A splash gate is raised and lowered to allow a robot to load and unload parts. The motor outputs to raise or lower the gate are controlled by pulsed input signals from the robot. There is a normally closed limit switch at t he bottom of the gate track, and normally open limit switches at the top. When the splash gate is down, the output indicator Gate Down will turn on. The output signal Gate Up Indicator will turn on when the gate is up.

The motor cannot change direction once it has been commanded to move.

Ignore any Blocks which names begin with Emulate.

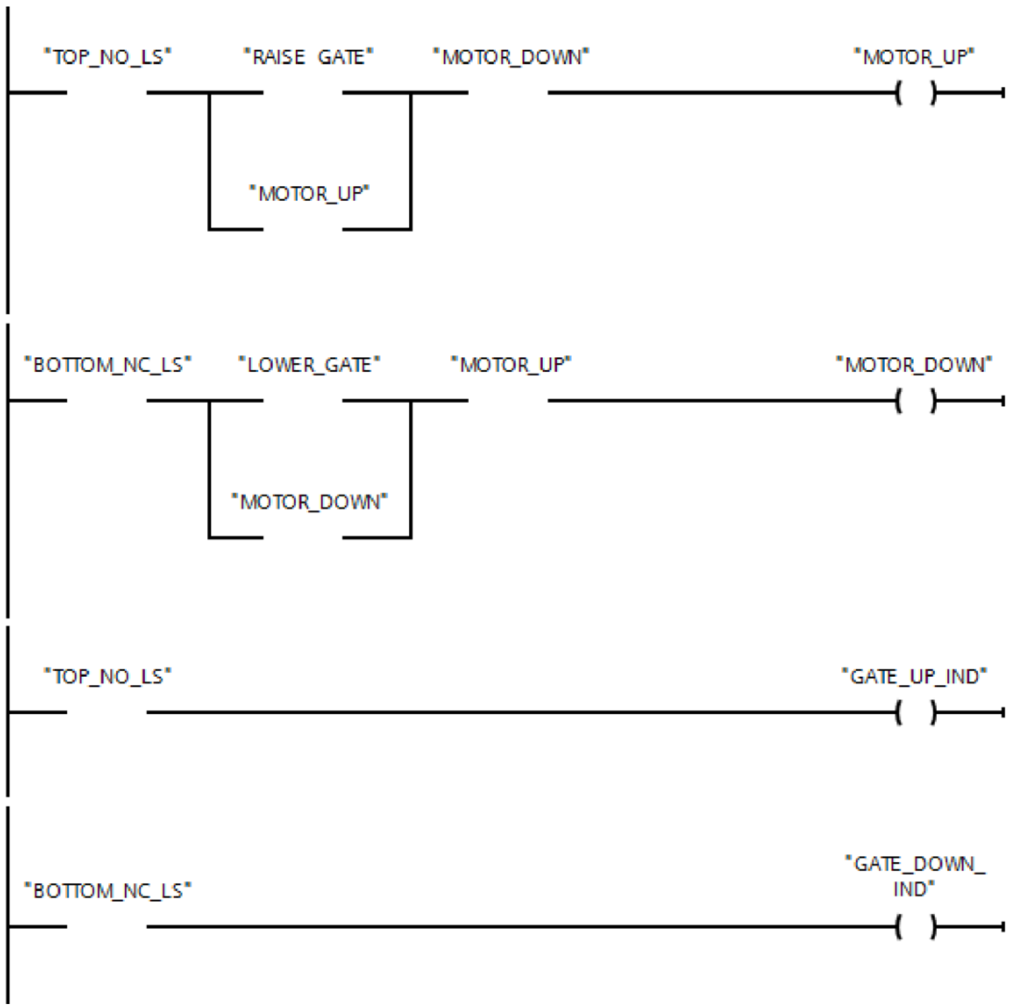
INPUTS:

Address	PLC Tag	Description
[redacted]	[redacted]	Raise Gate Signal from Robot
[redacted]	[redacted]	Lower Gate Signal from Robot
[redacted]	[redacted]	Bottom Normally Closed Limit Switch
[redacted]	[redacted]	Top Normally Open Limit Switch

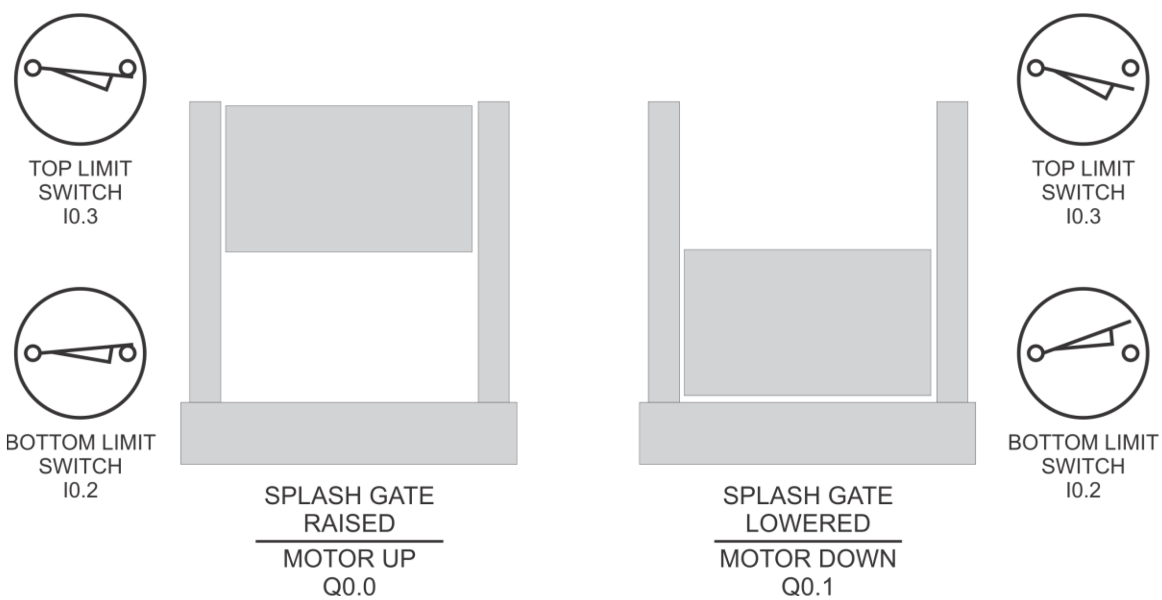
OUTPUTS:

Address	PLC Tag	Description
[redacted]	[redacted]	Up Motor Contactor
[redacted]	[redacted]	Down Motor Contactor
[redacted]	[redacted]	Gate Up Indicator
[redacted]	[redacted]	Gate Down Indicator

SEE THE NEXT PAGE FOR ILLUSTRATION.



NOTE: The limit switches are depicted in their <u>actual</u> positions.

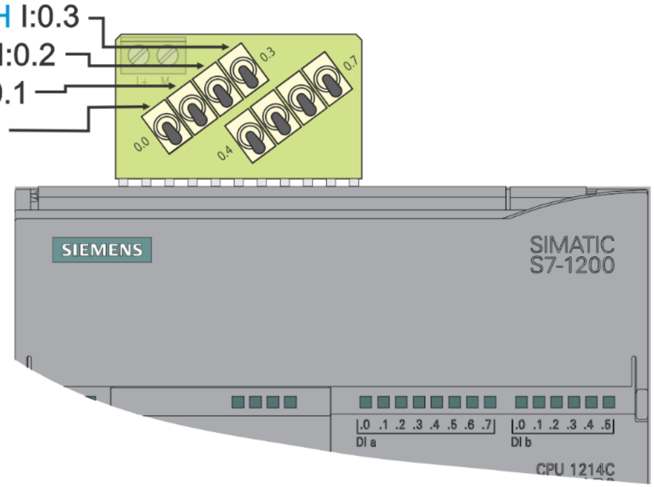


DOWNLOAD HMI FOR EXAM PROBLEM.

<u>SEQUENCE</u>

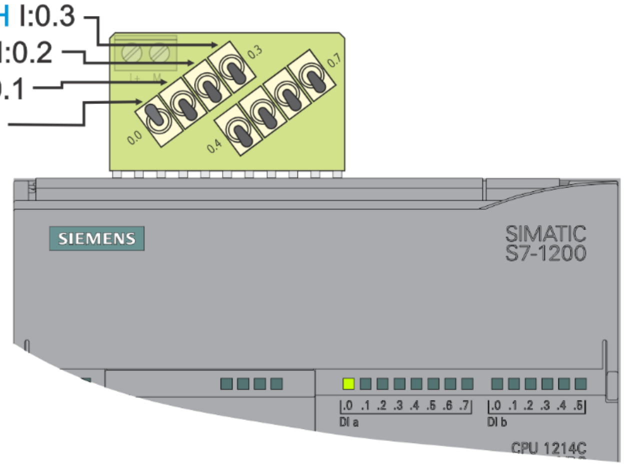
Begin with Gate in the Lowered Position.

Initial switch settings, push down **all** switches.



<u>RAISE GATE</u>

Push up the **RAISE GATE** switch.



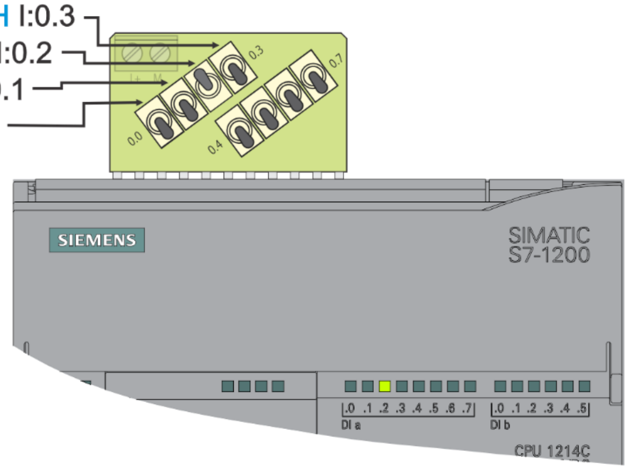
- - THEN - -

Push down the **RAISE GATE** switch.

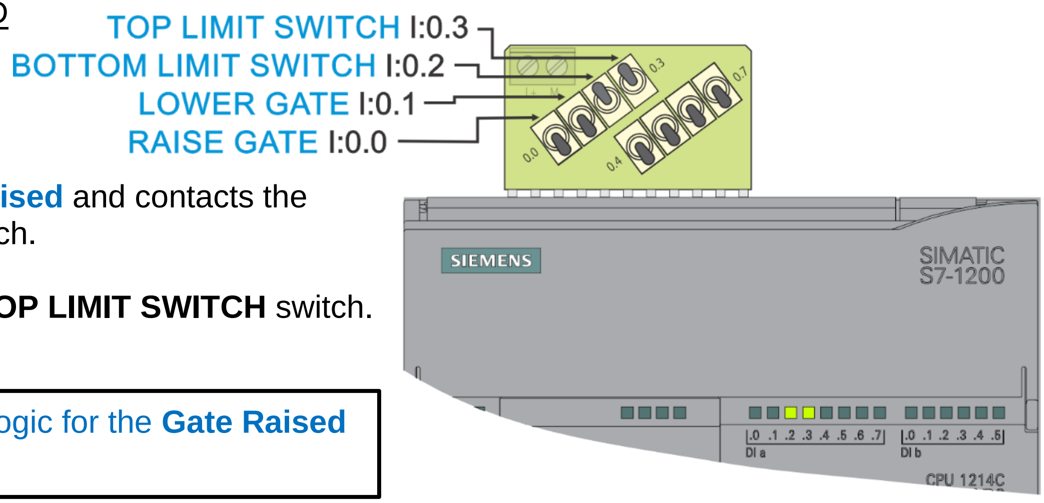
- - THEN - -

Gate is raising and moves off the Bottom Limit Switch.

Push up the **BOTTOM LIMIT SWITCH** switch.



GATE RAISED

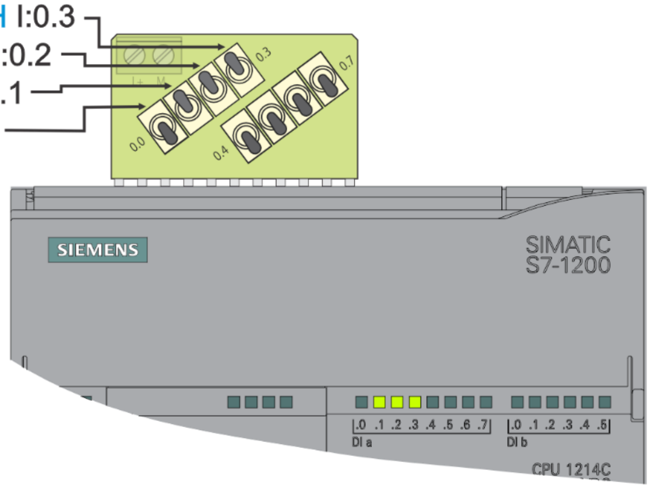


The gate is raised and contacts the Top Limit Switch.

Push up the **TOP LIMIT SWITCH** switch.

Check your logic for the **Gate Raised Indicator**.

LOWER GATE

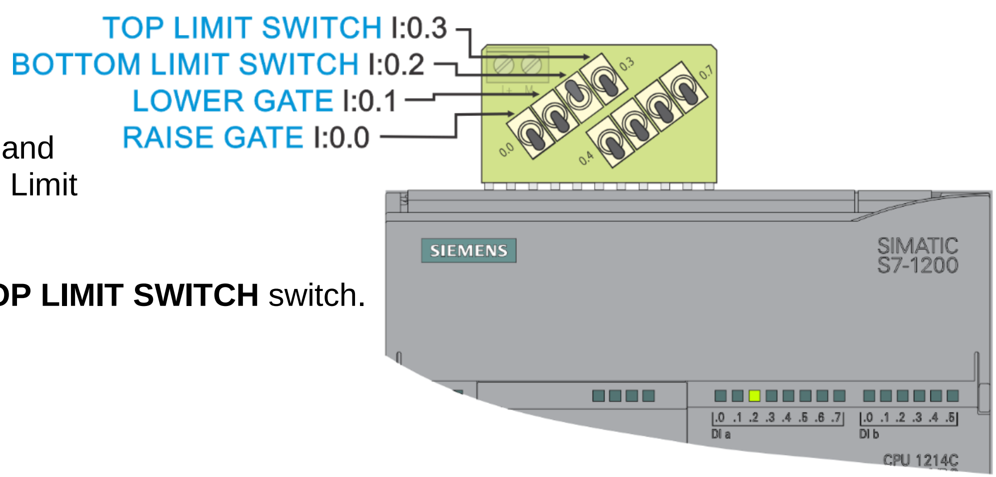


Push up the **LOWER GATE** switch.

- - THEN - -

Push down the **LOWER GATE** switch.

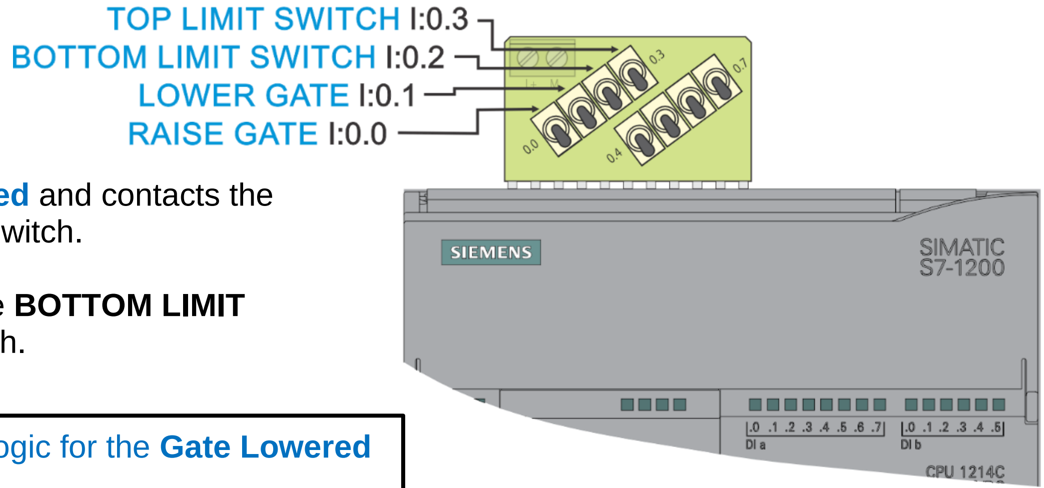
- - THEN - -



Gate is lowering and moves off the Top Limit Switch.

Push down the **TOP LIMIT SWITCH** switch.

GATE LOWERED



Gate is lowered and contacts the Bottom Limit Switch.

Push down the **BOTTOM LIMIT SWITCH** switch.

Check your logic for the **Gate Lowered Indicator**.

Remember to print to OneNote in the MIDTERM Tab as PROG3